

MONTE CARLO ASSIGNMENT 2

RUOLIN WANG, LEONARD

1. SELF-AVOIDING WALKS (SAWS) IN $\mathbb{Z}^d$

1.1. Question 1. Denote $c_k(d) = |S_k(d)|$, where $S_k(d)$ is the set of all possible SAWs of length k starting at the origin in $\mathbb{Z}^d$. $\mathbb{Z}^d$ is defined rigorously by:

$$\mathbf{S}_n(d) = \left\{ x_{0:n} \in \mathbb{Z}^{d(n+1)} : x_0 = \mathbf{0}, |x_k - x_{k-1}| = 1, x_\ell \neq x_k, \forall 0 \leq \ell < k \leq n \right\}.$$

Then consider the inequality

$$c_{n+m}(d) \leq c_n(d)c_m(d),$$

for fixed d and $n \leq 1, m \leq 1$. $c_{n+m}(d)$ is the number of all possible SAWs with length $n+m$ starting at the origin in $\mathbb{Z}^d$. For the right-hand side, we first take any SAWs with length n starting at the origin in $\mathbb{Z}^d$, there are $c_n(d)$ walks. Next, from the endpoint of that walk, if we do not consider the self-avoidance restriction, there are $c_m(d)$ walks with length m . Therefore, $c_{n+m}(d)$ must be at most $c_n(d)c_m(d)$.

1.2. Question 2. A sequence $(a_n)_{n \geq 1}$ is called *subadditive* if $a_{m+n} \leq a_m + a_n$. *Fekete's lemma* states that for every subadditive sequence $(a_n)_{n \geq 1}$, the limit $\lim_{n \rightarrow \infty} a_n/n$ exists and is equal to $\inf_{n \geq 1} a_n/n$ (which may be equal to $-\infty$). To use Fekete's lemma to prove the limit

$$\mu_d = \lim_{n \rightarrow \infty} c_n(d)^{1/n}$$

exists, observe first that a log transform is necessary in order for us to apply Fekete's lemma:

$$\log c_{n+m}(d) \leq \log c_n(d) + \log c_m(d).$$

This shows what the sequence $(a_k) = \log c_k(d)$ is subadditive. Then Fekete's lemma guarantees that the limit

$$\lim_{n \rightarrow \infty} \frac{\log c_n(d)}{n}$$

exists. Let $\log \mu_d$ be the limit above, then

$$\mu_d = \lim_{n \rightarrow \infty} e^{\frac{\log c_n(d)}{n}} = \lim_{n \rightarrow \infty} c_n(d)^{1/n}$$

which is exactly what we want to prove.

1.3. Question 3. We first estimate $c_n(d)$ using a 'naive' approach, that is, using the *sequential importance sampling* (SIS) algorithm with instrumental distribution g_n being that of a standard *random walk* $(X_k)_{k=0}^n$ in $\mathbb{Z}^2$, where $X_0 = \mathbf{0}$ and each X_{k+1} is drawn uniformly among the four neighbours of X_k . We set the random direction as: 1=up, 2=down, 3=left, 4=right with steps $[0,1], [0,-1], [-1,0], [1,0]$ respectively. Then draw from $\text{randi}(4)$ so that the instrumental distribution g_n is uniformly distributed. Then we check for each walk if no position is revisited and

Date: February 2025.

count the total number N_{SA} of such walks. Lastly, $c_n(2)$ is estimated by taking the ratio with total walks and multiplying by 4^n because there are in total 4^n of possible ways at each position:

$$\hat{c}_n(2) = \left(\frac{N_{SA}}{N} \right) \times 4^n$$

Below is the estimation (with 100000 samples) base on the 'naive' estimation of $c_n(2)$ for different values of n :

n	Estimated $\hat{c}_n(2)$	True $c_n(2)$
1	4.00 $\approx$ 4	4
2	11.99 $\approx$ 12	12
3	35.96 $\approx$ 36	36
4	100.05 $\approx$ 100	100
5	284.35 $\approx$ 284	284
6	782.13 $\approx$ 782	780
7	2168.91 $\approx$ 2169	2172
8	5919.87 $\approx$ 5920	5916
9	16342.06 $\approx$ 16342	16268
10	44522.54 $\approx$ 44523	44100

TABLE 1. Comparison of estimated and true values of $c_n(2)$ for different n

Note that the true $c_n(2)$ can be found in [1, Self-Avoiding Walks], a direct comparison indicates that the 'naive' SIS method works better for smaller n . For larger n this method is extremely inefficient as the SAW probability will drop exponentially. As the consequence, this 'naive' method leads to high variance and unreliable estimates.

1.4. Question 4. In this question, we tend to improve the accuracy of estimation by changing the instrumental distribution g_n from Question 3 to self-avoiding random walks. By lecture notes, c_n is estimated by:

$$c_n \approx \frac{1}{N} \sum_{i=1}^N \omega_N^i.$$

We choose the weight for a SAW of length n is the product of the number of available choices d_k at each step k . That is :

$$\omega_n^i = \prod_{k=1}^n d_k$$

This compensates for the non-uniform proposal distribution g_n and at each step k , the cumulative weight is updated as $w_k = w_{k-1} \cdot d_k$ because:

$$\omega_{n+1}^i = \omega_n^i \cdot \frac{c_n}{c_{n+1}} \cdot d_k$$

But since $c_{n+1} = c_n \cdot \mathbb{E}[d_k]$, the weight simplifies to

$$\omega_{n+1}^i = \omega_n^i \cdot d_k.$$

Note that if the path is trapped, meaning that there is no unvisited point, then we set the remaining weights to be 0. In this way, every sample contributes to the estimate (via weights), even if walks get trapped at later steps. Below is the estimation (with $N = 100000$ samples) based on the improved 'naive' estimation of $c_n(2)$ for different values of n :

n	Estimated $c_n(2)$	True $c_n(2)$
1	4	4
2	12	12
3	36	36
4	100	100
5	284	284
6	780	780
7	2171	2172
8	5911	5916
9	16262	16268
10	44092	44100
11	120151	120292
12	324752	324932
13	880829	881500
14	2370528	2374444
15	6402986	6416596
16	17226010	17245332
17	46364619	46466676
18	124318408	124658732

TABLE 2. Comparison of estimated and true values of $c_n(2)$ for different values of n .

Note that the true $c_n(2)$ can be found in [1, Self-Avoiding Walks]. The improved SIS method outperforms the naive approach by efficiently estimating $c_n(2)$ for larger n with lower variance, however the computational cost is still very high for larger n . Figure 1 shows a visualization of the estimates.

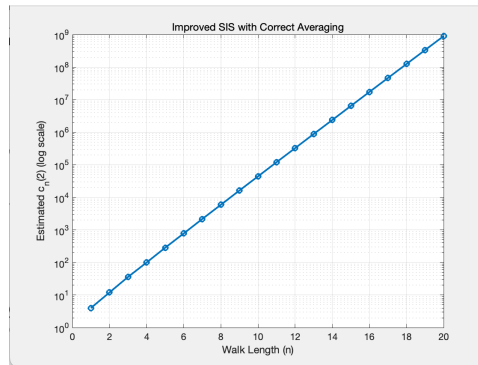


FIGURE 1. Estimated $c_n(2)$ with n from 1 to 20

1.5. **Question 5.** In this question we want to improve further the method in question 4 by using SISR. By lecture notes,

$$c_{N,n}^{\text{SISR}} = \prod_{k=0}^n \left(\frac{1}{N} \sum_{i=1}^N \omega_k^i \right).$$

At each step k we compute first $\bar{\omega}_k = \frac{1}{N} \sum_{i=1}^N \omega_k^i$, then update c_n as $c_n = \prod_{m=1}^k \bar{\omega}_m$, ensuring alignment with the theorem

$$c_{N,n}^{\text{SISR}} = \prod_{k=0}^n \bar{\omega}_k.$$

In the resampling stage, the samples are replaced with probabilities by the importance weights. Below shows our estimation with sample size $N = 100000$:

n	True $c_n(2)$	Estimated $c_n(2)$	Approximation
1	4	4.00	≈ 4
2	12	12.00	≈ 12
3	36	36.00	≈ 36
4	100	99.99	≈ 100
5	284	284.00	≈ 284
6	780	779.64	≈ 780
7	2172	2169.81	≈ 2170
8	5916	5913.63	≈ 5914
9	16268	16267.19	≈ 16267
10	44100	44134.03	≈ 44134
11	120292	120399.43	≈ 120399
12	324932	325463.41	≈ 325463
13	881500	882548.92	≈ 882549
14	2374444	2379748.30	≈ 2379748
15	6416596	6431965.43	≈ 6431965
16	17245332	17288449.76	≈ 17288450
17	46466676	46594882.68	≈ 46594883
18	124658732	124999889.61	≈ 124999890

TABLE 3. Comparison of True and Estimated $c_n(2)$ Values

After comparison between Table 2 and Table 3, we notice that for smaller n , both SIS and SISR estimations are accurate. However, for larger n values the SISR method has an increased variance lead to less accurate estimations.

1.6. **Question 6.** According to formula given:

$$c_n(2) \sim A_2 \mu_2^n n^{\gamma_2 - 1} \quad \text{as } n \rightarrow \infty.$$

To estimate the parameters we take the log on both sides:

$$\log(c_n) \approx \log(A_2) + n \log(\mu_2) + (\gamma_2 - 1) \log(n).$$

Let $y_n = \log(c_n)$, $\beta_0 = \log(A_2)$, $\beta_1 = \log(\mu_2)$ and $\beta_2 = \gamma_2 - 1$. Then the equation is in a standard linear regression form for us to estimate.

$$y_n = \beta_0 + \beta_1 n + \beta_2 \log(n).$$

Since the SIS method gives better estimation, we first make 3 sets of estimations from Question 4 and then use MATLAB's built-in backslash operator to solve the linear regression equation. To speed up the process we change the sample size $N = 10000$, below shows the estimation results in 3 different iterations:

Iteration	A_2	μ_2	γ_2
1	1.485116	2.655924	1.227829
2	1.483456	2.663432	1.219607
3	1.484322	2.661379	1.221269

TABLE 4. Estimated values of A_2 , μ_2 , and γ_2 over 3 iterations

Compare our estimations on the parameters with theoretical values we see that our estimations are relatively accurate. Amount all parameters, μ_2 has the least variations across iterations and γ_2 shows the largest variations among the three parameters. This is because there is an exponential n to μ_2 , making small variations in μ_2 have a much larger effect on $c_n(2)$. This makes it the easiest parameter to estimate. On the other hand, γ_2 is multiplied by $\ln(n)$ and is growing relatively slow with respect to n , this leads to a larger estimation uncertainty.

1.7. **Question 7.** The following inequality is to be verified

$$(1.1) \quad d \leq \mu \leq 2d - 1$$

for a general d . To see its validity, the following inequality [1, Self-Avoiding Walks] for an n -step SAW will be used

$$(1.2) \quad d^n \leq c_n \leq (2d)(2d - 1)^{n-1}.$$

The upper bound in the above inequality is explained by having $2d$ choices to move to from the starting point and then having $2d - 1$ choices for each remaining $n - 1$ steps, given that the walk never encounters previously visited points as neighbors again (other than the last one visited). As such, it is clearly the biggest possible set of n -step walks, such that the size of this set is an upper bound for the number of c_n -walks. The lower bound is given by the size of the set of n -step walks which move only in the positive (or negative) direction. From inequality (1.2) we obtain

$$d \leq c_n^{1/n} \leq (2d)^{1/n} (2d - 1)^{-1/n} (2d - 1).$$

Taking the limit yield (1.1), the desired inequality.

1.8. **Question 8.** Now, the following inequality is to be verified

$$(1.3) \quad A_d \geq 1, \quad \text{for } d \geq 5.$$

It is known (from the assignment instructions) that

$$(1.4) \quad c_n \sim A_n \mu_n^d n^{\gamma_d - 1}, \quad \text{for } d \geq 5,$$

with $\gamma_d = 1$ for $n \geq 5$. Thus

$$(1.5) \quad c_n \sim A_n, \quad \text{for } d \geq 5.$$

holds. Using (1.5) in conjunction with

$$c_{n+m}(d) \leq c_n(d) c_m(d),$$

(which was shown to hold true in task 1), will give the desired result. Thus, using submultiplicativity of c_n together with (1.5) gives

$$(1.6) \quad A_d \mu_d^{n+m} \leq A_d \mu_d^n A_d \mu_d^m = A_d^2 \mu_d^{n+m} \implies 1 \leq A_d, \quad \text{for } d \geq 5,$$

which was to be shown. (Note that μ_d^n was treated as a constant in (1.6), as it is known from task 2 that the limit exists given any d .)

1.9. Question 9. In this task the SISR method is used once again as in task 6. This time the method is used to estimate A_d, μ_d and γ_d for some $d \geq 3$. The estimates will then be compared with the bounds given by inequalities (1.1) and (1.3). Then an estimate for μ_d will be compared with the asymptotic bound given by

$$(1.7) \quad \mu_d \sim 2d - 1 - 1/2d - 3/(2d)^2 - 16/(2d)^3 + O(1/d^4).$$

Once again, the sample size is chosen to be $N = 10000$ when computing the estimates. In this task, the dimension is chosen as $d = 5$. Thus, some changes are made in the MATLAB code to incorporate 5 dimensions instead of 2.

With SISR estimates of $c_n(5)$, the following estimates of A_5, μ_5 and γ_5 are obtained

$$\hat{A}_5 = 1.130941$$

$$\hat{\mu}_5 = 8.829618$$

$$\hat{\gamma}_5 = 1.034828.$$

The bound given by equation (1.1) is verified below

$$d = 5 \leq \hat{\mu}_5 = 8.829618 \leq 2 \times 5 - 1 = 9.$$

To verify inequality (1.3), note that

$$\hat{A}_5 = 1.130941 \geq 1.$$

Finally, comparing an estimated μ_{10} (again, having made adjustments to the MATLAB code) with the bound given by (1.6), note that

$$\hat{\mu}_{10} = 18.931655 \sim 2 \times 10 - \frac{1}{2 \times 10} - \frac{3}{(2 \times 10)^2} - \frac{16}{(2 \times 10)^3} = 18.9405.$$

Thus, the different bounds have been checked and verified with $d = 5$ for (1.1) and (1.3), and $d = 10$ for (1.7).

In conclusion, given that the estimates followed the given bounds, the SISR method is a good way to estimate the parameters that are used to describe the asymptotic bounds for different lengths of SAW with different dimensions.

2. FILTER ESTIMATION OF NOISY POPULATION MEASUREMENTS

In the second part of the assignment, a stochastic process $\{X_k\}_{k=1}^N$ documents the relative population sizes of some organism at generation k . A simplified relative population model has the following mechanics

$$X_{k+1} = R_{k+1} X_k (1 - X_k), \quad R_{k+1} \in U(0.8, 3.8), \text{ i.i.d., } k = 0, 1, 2, \dots,$$

where R_k are stochastic reproduction rates. It is assumed that $X_0 \in U(0.6, 0.99)$. The exact values of the X_k 's are unknown, however, the pair (Y_k, X_k) constitutes

a Hidden Markov Model for which

$$Y_k|X_k = x \in U(0.8, 1.25)$$

holds.

2.1. **a).** In this part the filter expectation $E[Y_k|X_{0:k}]$ for $k = 0, 1, 2, \dots, 100$ with the number of particles $N = 500, 1000, 10000$ is to be estimated. In other words, the hidden relative population sizes at each generation k will be estimated by using the "noisy" observations Y_k . To do this the code in Lecture notes 7 page 24 will be modified to fit the pertaining task. In addition, estimates are made given three different N 's.

It is seen in Figure 1 through 3 that the difference between the different N -sizes is barely noticeable, especially comparing $N = 1000$ and $N = 10000$. However, since the computational work is not very heavy, choosing the larger particle size $N = 10000$ is a reasonable choice.

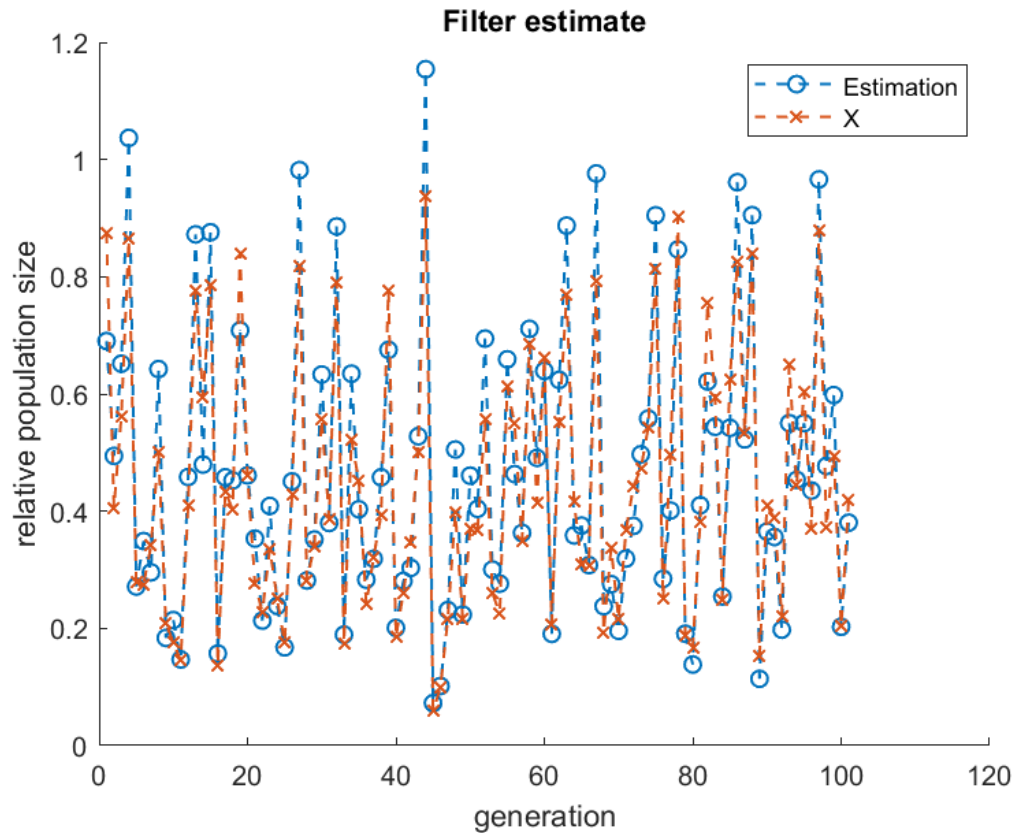


FIGURE 2. Filter estimates (blue, circles) vs true values (red, crosses), with particle size $N=500$.

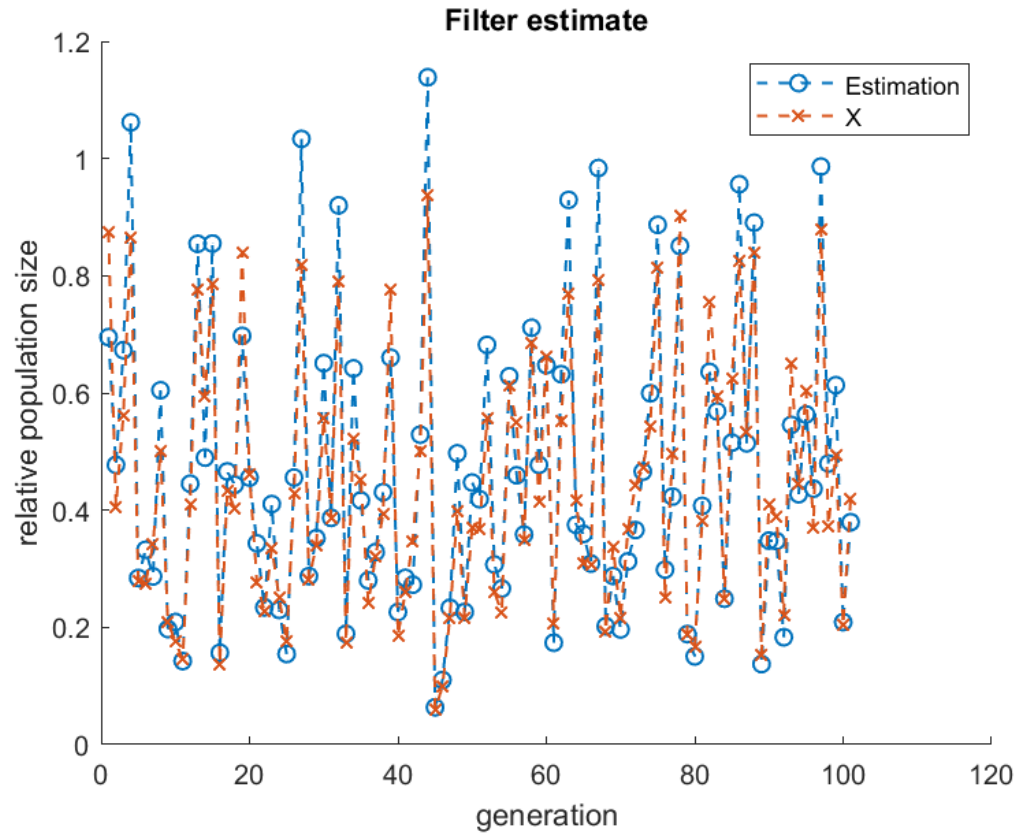


FIGURE 3. Filter estimates (blue, circles) vs true values (red, crosses), with particle size $N=1000$.

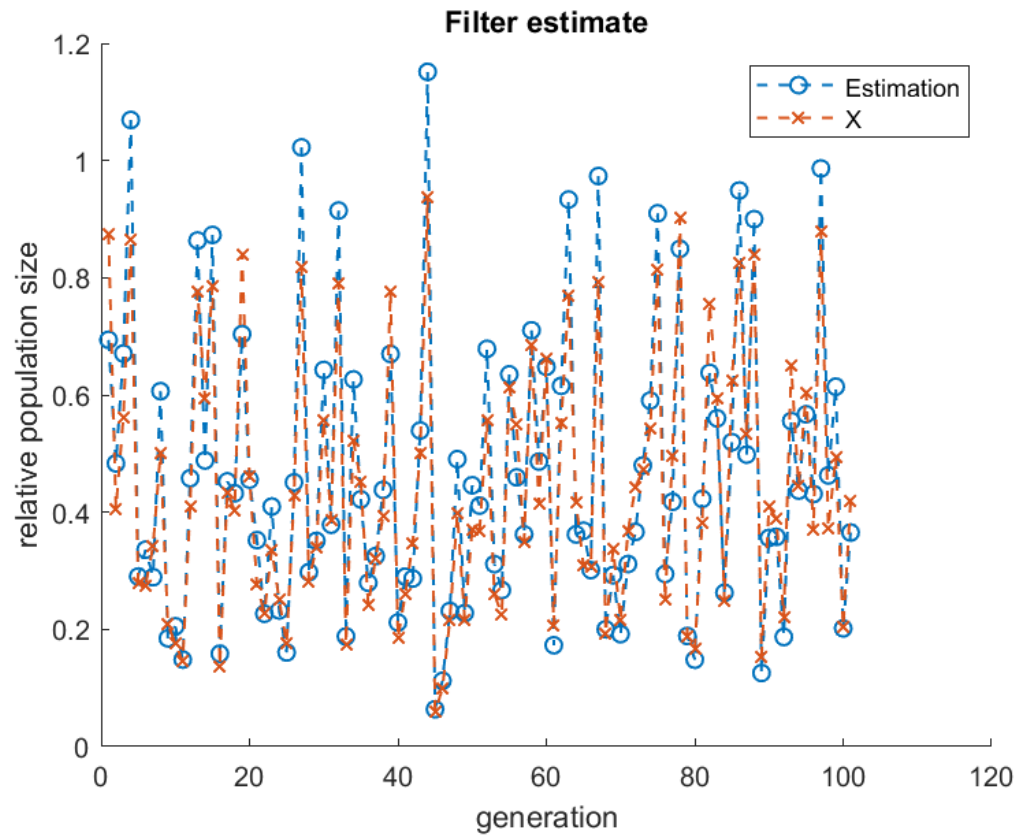


FIGURE 4. Filter estimates (blue, circles) vs true values (red, crosses), with particle size $N=10000$.

2.2. **b).** In this section point wise confidence intervals for X_k are made for each generation $k = 0, 1, 2, \dots, 100$ for the three different $N = 500$, $N = 1000$ and $N = 10000$. To do this, the code in Lecture note 7 page 25 is modified to fit the current setting. Again, letting $N = 10000$ seem to be the most reasonable choice, as the computational work for the computer is not very large. Tables containing the confidence intervals from generation 1 to 30 for $N = 500, 1000, 10000$ are given below. The tables also contain information pertaining section a) above, i.e., the estimated value compared to the real value X_k . In the tables (for one given simulation) no substantial differences between the three different choices of N can be seen. To notice any difference for the different sizes of N , several simulations has to be made and compared against each other.

Results for N = 500				
Generation	Estimate	True Value	Confidence Interval	Inside CI
1	0.6903	0.8744	[0.6016, 0.7902]	0
2	0.4943	0.4059	[0.3949, 0.5963]	1
3	0.6517	0.5618	[0.5355, 0.8173]	1
4	1.0369	0.8642	[0.8630, 1.3050]	1
5	0.2714	0.2782	[0.2325, 0.3354]	1
6	0.3490	0.2752	[0.2831, 0.4114]	0
7	0.2955	0.3418	[0.2397, 0.3586]	1
8	0.6424	0.5011	[0.4861, 0.7393]	1
9	0.1837	0.2084	[0.1540, 0.2235]	1
10	0.2142	0.1762	[0.1747, 0.2575]	1
11	0.1469	0.1454	[0.1218, 0.1727]	1
12	0.4584	0.4103	[0.3844, 0.5512]	1
13	0.8723	0.7772	[0.6834, 1.0437]	1
14	0.4795	0.5951	[0.3919, 0.5994]	1
15	0.8759	0.7859	[0.6969, 1.0541]	1
16	0.1571	0.1368	[0.1276, 0.1987]	1
17	0.4578	0.4332	[0.3724, 0.5579]	1
18	0.4524	0.4028	[0.3451, 0.5357]	1
19	0.7083	0.8402	[0.5750, 0.8636]	1
20	0.4611	0.4603	[0.3693, 0.5630]	1
21	0.3532	0.2765	[0.2788, 0.4299]	0
22	0.2133	0.2280	[0.1810, 0.2701]	1
23	0.4088	0.3362	[0.3265, 0.5063]	1
24	0.2383	0.2500	[0.1886, 0.2892]	1
25	0.1676	0.1756	[0.1324, 0.1911]	1
26	0.4507	0.4280	[0.3779, 0.5598]	1
27	0.9819	0.8189	[0.8199, 1.2456]	0
28	0.2817	0.2821	[0.2327, 0.3613]	1
29	0.3515	0.3394	[0.2852, 0.4317]	1
30	0.6332	0.5575	[0.5185, 0.7765]	1

Table 5. Estimates, true values, confidence intervals and whether the true value is in the confidence interval (1 for yes and 0 for no) for generation 1 to 30 for N=500.

Results for N = 500				
Generation	Estimate	True Value	Confidence Interval	Inside CI
1	0.6952	0.8744	[0.6054, 0.7900]	0
2	0.4771	0.4059	[0.3887, 0.5923]	1
3	0.6730	0.5618	[0.5447, 0.8193]	1
4	1.0618	0.8642	[0.8558, 1.3100]	1
5	0.2843	0.2782	[0.2322, 0.3570]	1
6	0.3325	0.2752	[0.2752, 0.4163]	0
7	0.2869	0.3418	[0.2417, 0.3499]	1
8	0.6045	0.5011	[0.4832, 0.7357]	1
9	0.1971	0.2084	[0.1629, 0.2272]	1
10	0.2091	0.1762	[0.1781, 0.2557]	0
11	0.1425	0.1454	[0.1184, 0.1755]	1
12	0.4451	0.4103	[0.3714, 0.5433]	1
13	0.8546	0.7772	[0.6906, 1.0601]	1
14	0.4896	0.5951	[0.4054, 0.5995]	1
15	0.8550	0.7859	[0.6908, 1.0706]	1
16	0.1567	0.1368	[0.1273, 0.1940]	1
17	0.4661	0.4332	[0.3788, 0.5531]	1
18	0.4434	0.4028	[0.3573, 0.5365]	1
19	0.6974	0.8402	[0.5689, 0.8449]	1
20	0.4556	0.4603	[0.3670, 0.5544]	1
21	0.3434	0.2765	[0.2829, 0.4242]	0
22	0.2341	0.2280	[0.1815, 0.2757]	1
23	0.4104	0.3362	[0.3419, 0.4918]	0
24	0.2298	0.2500	[0.1880, 0.2773]	1
25	0.1546	0.1756	[0.1300, 0.1892]	1
26	0.4557	0.4280	[0.3643, 0.5548]	1
27	1.0335	0.8189	[0.8351, 1.2640]	0
28	0.2875	0.2821	[0.2373, 0.3501]	1
29	0.3522	0.3394	[0.2813, 0.4333]	1
30	0.6509	0.5575	[0.5252, 0.7809]	1

Table 6. Estimates, true values, confidence intervals and whether the true value is in the confidence interval (1 for yes and 0 for no) for generation 1 to 30 for N=1000.

Results for N = 10000				
Generation	Estimate	True Value	Confidence Interval	Inside CI
1	0.6941	0.8744	[0.6047, 0.7907]	0
2	0.4832	0.4059	[0.3896, 0.5969]	1
3	0.6707	0.5618	[0.5354, 0.8217]	1
4	1.0691	0.8642	[0.8613, 1.3174]	1
5	0.2903	0.2782	[0.2327, 0.3562]	1
6	0.3359	0.2752	[0.2706, 0.4147]	1
7	0.2890	0.3418	[0.2321, 0.3546]	1
8	0.6064	0.5011	[0.4853, 0.7430]	1
9	0.1847	0.2084	[0.1493, 0.2270]	1
10	0.2053	0.1762	[0.1664, 0.2540]	1
11	0.1484	0.1454	[0.1180, 0.1826]	1
12	0.4578	0.4103	[0.3670, 0.5610]	1
13	0.8635	0.7772	[0.6930, 1.0491]	1
14	0.4881	0.5951	[0.3920, 0.5956]	1
15	0.8731	0.7859	[0.6994, 1.0677]	1
16	0.1585	0.1368	[0.1281, 0.1957]	1
17	0.4525	0.4332	[0.3630, 0.5538]	1
18	0.4322	0.4028	[0.3465, 0.5332]	1
19	0.7037	0.8402	[0.5648, 0.8626]	1
20	0.4555	0.4603	[0.3662, 0.5577]	1
21	0.3522	0.2765	[0.2804, 0.4277]	0
22	0.2267	0.2280	[0.1787, 0.2746]	1
23	0.4097	0.3362	[0.3292, 0.5055]	1
24	0.2322	0.2500	[0.1883, 0.2875]	1
25	0.1606	0.1756	[0.1280, 0.1961]	1
26	0.4509	0.4280	[0.3655, 0.5559]	1
27	1.0225	0.8189	[0.8206, 1.2539]	0
28	0.2976	0.2821	[0.2394, 0.3587]	1
29	0.3511	0.3394	[0.2877, 0.4333]	1
30	0.6427	0.5575	[0.5162, 0.7879]	1

Table 7. Estimates, true values, confidence intervals and whether the true value is in the confidence interval (1 for yes and 0 for no) for generation 1 to 30 for N=10000.

REFERENCES

- [1] G. Slade, “Self-Avoiding Walks,” 1994. Available at: <https://personal.math.ubc.ca/~slade/intelligencer.pdf>.